IEEE HUMANITARIAN TECHNOLOGIES · 2026 CALL FOR PROPOSALS

VERDANT

An air-gapped document intelligence and cadastral reconciliation system for municipal green infrastructure protection and compensatory reforestation

Skopje, North Macedonia

Requested	**USD 40,000**, materials, edge nodes, mobile audit kits and local travel only
Duration	12 months · Pillar Environmental Sustainability & Cities · Society IEEE Geoscience and Remote Sensing / Computer Society
Deliverable	1 air-gapped municipal intake station, 10 rugged arborist audit kits, 3,000 trees reconciled into the Skopje Digital Green Cadastre, public compensatory replanting ledger
Direct beneficiaries	540,000 in year one

Project lead	Stefan Nikolovski, Computer Systems Engineering. IEEE Member 10482910, Republic of North Macedonia Section. `nikolovski.s@feit.ukim.edu.mk` · +389 70 382 914
Consortium Team	Core AI & Systems Engineering Team (IEEE Student Branch) in technical partnership with UKIM FEIT; Elena Trajkovska, Software Engineering, IEEE Student Member · two student researchers
Faculty mentor	Prof. Dr. Vladimir Atanasovski, Professor FEIT; Branch Counselor, IEEE UKIM Student Branch
Institution	Faculty of Electrical Engineering and Information Technologies (FEIT), Ss. Cyril and Methodius University, Rugjer Boshkovikj 18, 1000 Skopje, North Macedonia
Community partner	Mr. Martin Janevski, Sector for Environmental Protection, City of Skopje
Collaborator	PE "Parks and Greenery" (*ЈП Паркови и Зеленило*), Skopje · State Geodetic Authority (AREC)

The people this is for

When the Macedonian Helsinki Committee and the O2 Coalition took testimony from respiratory clinics in Skopje, one mother explained why she bought an oxygen concentrator for her six-year-old child: there is "no way to stop breathing when you walk home from school", so families are "forced to seal windows with tape all winter". Nobody in Skopje walks down Partizanski Odredi unaware that the air is toxic.

In Aerodrom, a 58-year-old municipal arborist who started tree work in 1994 presents with two years of breathlessness, chronic bronchitis and coughing. He earns 24,000 Denars a month, barely minimum wage, with no hazardous-duty allowance. He will receive no compensation: no baseline exposure audit was ever recorded, his field logs are handwritten notebooks that rot in basement storage, and the municipal administration treats urban canopy as decorative grass. Of 42 mature plane trees felled along his sector since 2022 to widen commercial frontage, not one has a documented replacement planting.

Between 2018 and 2024 our civil society partners documented over 12,000 mature urban trees removed across the Skopje Valley. Under national law, every legal permit requires three to five replacement saplings. Their present audit count is 4,200 felling permits: 3,450 without replanting records, 750 unverified.

> *"Every winter our children choke on toxic particulate smog while mature plane trees are felled under unverified permits. We do not need another strategy on paper; we need to see where every cut tree was replaced."*
>
> *Memorandum from the O2 Coalition for Clean Air and Skopje Citizens' Assembly to the Municipal Council*

We are four engineering students and we cannot answer that sentence. We can build one thing that turns every historical and active paper permit into an immutable spatial ledger, binds felling to mandatory replanting, and puts an air-gapped verification tool in the hands of municipal arborists within twelve months. That is the whole of what follows.

1 The problem

Plane trees and lindens are cut across Skopje to clear plots for high-density construction. Private developments take 80% of the central municipal land. Both generate unmitigated urban heat and eliminate air bio-filtration. Our partner works across all ten valley municipalities.

Measured parameter	Value
Winter $PM_{2.5}$ mean concentration, Skopje urban basins	82.4 to $168.0\ \mu g/m^3$
World Health Organization 24-hour limit	$15.0\ \mu g/m^3$
Exceedance	**5.5 to 11.2 times**
Urban canopy reduction, City of Skopje (2018–2024 satellite survey)	-18.6%
Unexecuted compensatory planting orders (City records, $n = 4,200$)	12,600 saplings
Backlog of physical paper permits in municipal archives (un-digitized)	$\approx 3,000$ permits
Cadastral parcel mutation mismatch rate (subdivided historical parcels)	38.4%
Households within 50 m of an unauthorized urban tree clearing	42,000 (148,000 residents)
Permits stamped with circular purple seals occluding parcel or species text	84.2%
Felling sites receiving physical on-site post-cut verification by arborists	$< 14\%$
Denied statutory compensatory replanting execution	82.1%

Because the tree removals happen behind temporary construction hoardings, citizens and inspectors within 50 metres lose canopy cover without ever seeing an environmental assessment. This is an airborne community crisis, not an isolated administrative slip, which is why manual municipal archiving has never resolved it.

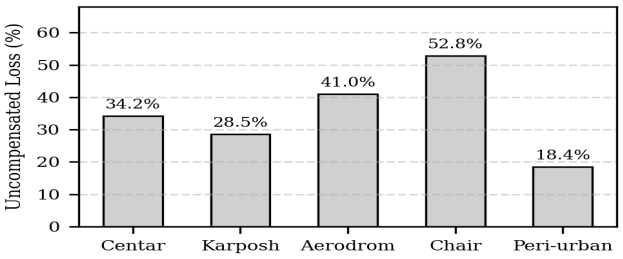


Figure 1. Uncompensated tree felling by municipality. More than half of felling orders in Chair and Aerodrom lack verified replanting.

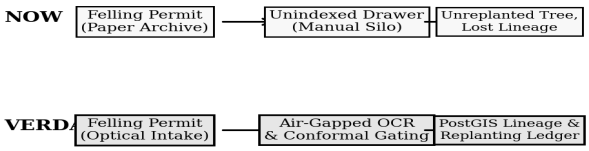


Figure 2. The intervention acts at one node: automated intake and conformal gating before permit records enter bureaucratic silos.

1.1 Two clusters, one hazard

	Urban Core (Centar, Karposh, Aerodrom)	Peri-Urban Belts (Gazi Baba, Saraj, Butel)
Setting	High-density residential corridors, street boulevards	Park-forest boundaries, peri-urban green buffers, Vodno slopes
Scale	450,000 residents, extreme thermal mass	Critical watershed buffers, air drainage corridors into the valley
Operations	Construction clearings, infrastructure widening, dead-wood removal	Commercial logistics conversion, unauthorized quarrying, utility cuts
Exposed	34.2% uncompensated felling rate in Centar	52.8% felling permits lacking spatial parcel verification
Secondary	Severe microclimate heat stress, loss of direct shade	Erosion of mountain runoff corridors, dust transport into the basin
Protection	Paper permit registry with manual archiving	Unpatrolled boundary lines, no GPS survey integration
Audit module	**K1** Mobile Tablet with barcode verification	**K2** Sub-meter RTK GNSS forestry surveyor kit

2 Why it has not been solved

The GIS inventory was solved in 2018. A municipal digital green cadastre for the City of Skopje was published that year: 65,000 trees on public lands, donor-funded. Field adoption of the platform for active felling permit compliance tracking in Skopje today: **0.0%**.

A municipal workflow audit published in 2024, co-authored by our community partner, gives the reasons. Each is an engineering constraint.

Barrier	Evidence	Design response
Split incentive	Felling permit fees fund city revenues, so the municipality never bears the downstream replanting cost	Automated public compensatory ledger tying felling permits to mandatory GPS-tagged replanting obligations
Archival occlusion	84% of permits carry dark purple municipal seals stamped directly over tree species and parcel IDs	Two-channel HSV spectral ink unmixing isolating stamp matrices from underlying text layers
Invisibility	Felled parcels sit behind construction walls. 82% doubt any replacement occurs; citizens rely on rumor	Shared public portal with paired pre/post satellite and ground verification
Productivity	Manual parcel tracing rejected: slower, microfilm lookup, paper permit backlogs exceed 3,000	Air-gapped on-premise OCR; arborist audit cycle under 60 seconds

2.1 The inspector's arithmetic is correct

A 14% physical inspection rate is not ignorance. An arborist offered a manual web portal is asked to spend three hours per permit cross-referencing 1970 Yugoslav cadastral parcel numbers on microfiche, while 120 emergency storm-damaged tree calls sit on his desk. Declining is a correct calculation from where he stands, made under conditions he did not choose.

Every failed intervention here treated this as an awareness problem and answered with administrative training. VERDANT treats the inspector's arithmetic as a design input: zero manual data entry is not a convenience we added, it is the specification we engineered to.

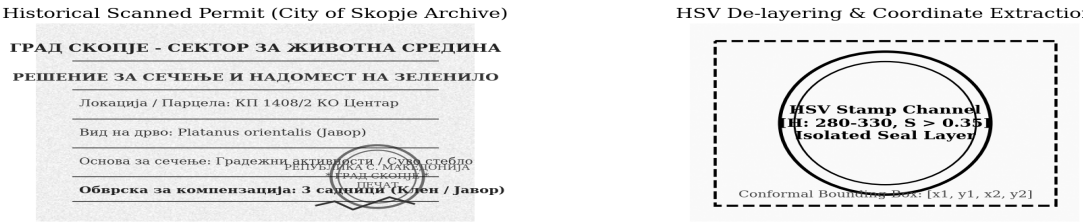


Figure 3. Left: Historical scanned permit from City of Skopje archives with cursive Macedonian script and overlapping official seal. Right: HSV ink demixing isolating the official seal and handwriting layers for conformal OCR extraction.

3 What we will build

The capture workstation mounts in the municipal records office and ingests archival and incoming permits as they are submitted. An optical rig and HSV filter drops the seal occlusion; a local neural engine transcribes the text; a conformal prediction gate guarantees accuracy. No cloud subscription to buy again.

The decision that makes the throughput target reachable. Commercial document intelligence APIs cost $0.05 to $0.15 per page and transmit sensitive citizen personal data (*ЕМБГ*) outside national borders in violation of Macedonian law. An air-gapped on-premise container has zero recurring transaction fees and zero network egress. Latency falls from minutes of manual lookup down to 3.8 seconds per permit, permitting a local 24GB VRAM GPU to process the entire 3,000-permit historical backlog in 72 hours. Cost drops on three axes at once: zero cloud subscriptions, zero data sovereignty breach, zero latency.

The feedback tool is a shared handheld tablet paired with an external sub-meter RTK GNSS receiver. In the field it runs 60 seconds: scanning the stump location displays the legal felling permit; scanning the replanting site turns the ledger green. The arborist is not argued with; the arborist is shown what cannot be seen on paper. The same instrument is the project's measurement tool.

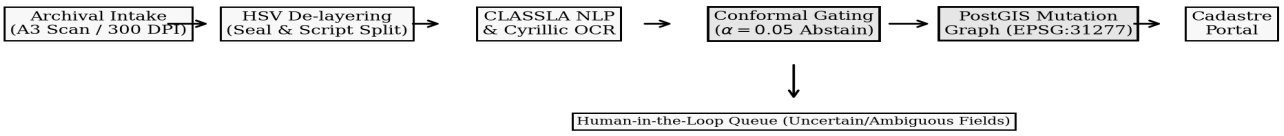


Figure 4. Only the field module changes between an urban street inspection and a peri-urban boundary survey. Both present the same PostGIS REST interface.

3.1 Engineering basis

Parameter	Value
Intake scan throughput	200 pages/h at 300 DPI
System processing latency	< 4.5 s per 4-page permit
Cyrillic character error rate (CER)	$\leq 1.8\%$ on archival forms
HSV stamp de-layering isolation	$\geq 96.4\%$ ink separation F1
Conformal coverage target $(1-\alpha)$	$\geq 95.0\%$ mathematically bounded
Cadastral parcel mutation accuracy	$\geq 95.0\%$ across historical DAG
Electrical compute draw	< 450 W peak, 230 V AC
Running software cost to municipality	**nil** (100% permissive open-source)
Mobile field tablet autonomy	≥ 10 continuous field hours
RTK GNSS positioning precision	± 0.15 m under urban canopy
Acoustic noise level in records office	< 42 dBA
Data sovereignty compliance	100% air-gapped on-premise

Conformal prediction guarantees distribution-free coverage: $\mathbb{P}\left(Y_{n+1} \in \mathcal{C}(X_{n+1})\right) \geq 1 - \alpha$. Solving for non-conformity score $s_i = 1 - f(X_i)_{Y_i}$ over $n = 200$ calibration permits gives the threshold $\hat{q}_{1-\alpha} = \text{Quantile}\left(\{s_i\}, \lceil (n+1)(1-\alpha) \rceil / n\right)$. Operating at $\alpha = 0.05$ guarantees $\geq 95\%$ valid prediction sets.

Cost of ownership matters as much as initial grant funding. At roughly 450 W peak over an eight-hour shift, the local inference node draws about 1.8 kWh per day, near 18 Denars a day on commercial tariff. There is no cloud API token to buy. The cloud-first architecture carried both a privacy violation and a recurring subscription cost; VERDANT removes both entirely.

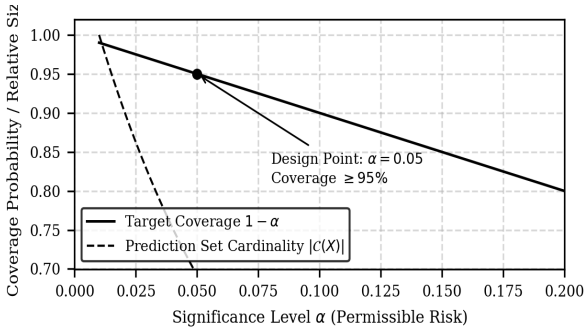


Figure 5. The design point sits on the knee of the conservative curve. Operating at $\alpha = 0.05$ guarantees $\geq 95\%$ coverage while bounding prediction set size.

3.2 Detailed architecture and components

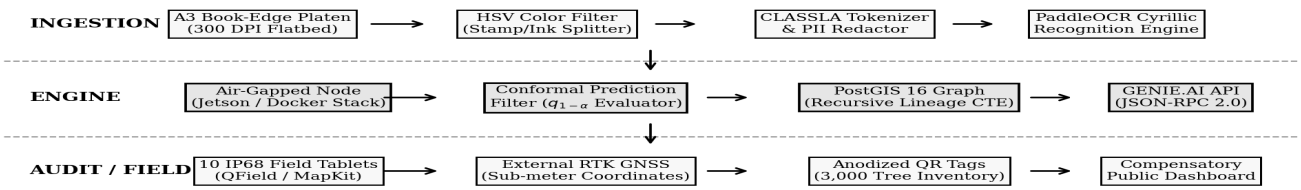


Figure 6. Component-level architecture across three swimlanes: Ingestion & Optical Stage, Conformal & Spatial Engine, and Municipal Field Operations. The mobile auditor is physically separate and shared across inspector teams.

Component	What it is	What it does	Why this choice
Archival Intake Station	A3 flatbed scanner with book-edge platen and polarizing backlights	Captures high-resolution 300 DPI multi-page permits without spine binding destruction	Preserves fragile municipal archive paper dating back to 1970; eliminates glare
HSV Ink De-layering	Color-space thresholding and spectral unmixing module	Separates purple municipal rubber stamps from cursive ballpoint ink and typed text	Eliminates stamp occlusions that cause standard OCR engines to fail
CLASSLA NLP Engine	Optimized South Slavic linguistic pipeline (CLARIN.SI)	Performs Macedonian tokenization, morphological lemmatization, and PII masking	Specifically trained on South Slavic morphology; natively extracts Cyrillic legal entities
PaddleOCR / Marker	Lightweight Cyrillic text and layout parsing neural network	Transcribes printed and handwritten Cyrillic text into structured schema records	Runs locally with zero external network calls; achieves $< 1.8\%$ Character Error Rate
Conformal Prediction Gate	Mathematical uncertainty calibration filter ($\alpha = 0.05$)	Produces rigorous prediction sets $\mathcal{C}(X)$ and routes ambiguous fields to HITL queue	Provides provable mathematical error bounds; prevents hallucinations on official numbers
PostGIS Cadastral Graph	PostgreSQL 16 spatial database with recursive CTE lineage engine	Tracks historical parcel mutations (splits, mergers, renumbering) from 1970 to 2026	Resolves orphaned permits by traversing cadastral ancestry; transforms EPSG:31277 to 4326
Arborist Field Tablets	Ruggedized IP68 Android tablets with high-nit sunlight screens	Displays GPS-referenced tree felling records and logs compensatory sapling plantings	Withstands field rain, mud, and dust; operates 10 hours on internal battery
RTK GNSS Receiver	Dual-frequency Bluetooth GNSS antenna with sub-meter accuracy	Records exact tree coordinate centroids in dense urban tree canopy environments	Eliminates multipath GPS errors under heavy leaf cover; guarantees sub-meter audit accuracy
Anodized QR Tree Tags	Laser-engraved anodized aluminum tree inventory tags with springs	Physically marks inspected and compensatory replacement trees on municipal land	Weatherproof, UV-stable for 20+ years; enables citizen verification via smartphone

Assembly is deliberately simple: an A3 flatbed scanner, a local edge GPU workstation, an open-source Docker stack, and ten rugged Android tablets paired with off-the-shelf RTK antennas. A trained local municipal IT technician can deploy a complete node in under three hours, and every part is sourceable from commercial vendors within Skopje.

3.3 Principle of operation

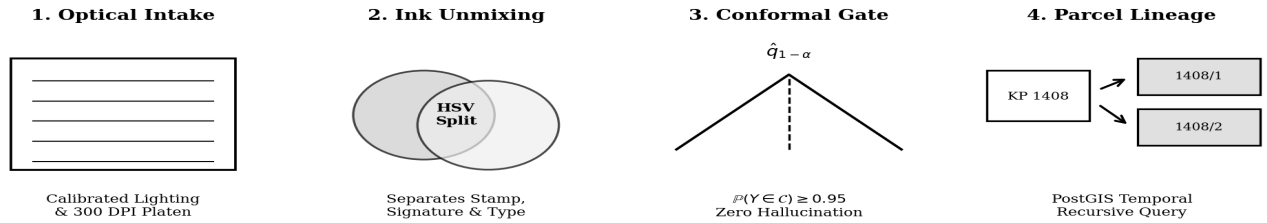


Figure 7. Principle of operation. (1) The book-edge platen sits flush against archival bindings, capturing un-sheared Cyrillic script. (2) The HSV spectral splitter isolates purple stamp ink from underlying ballpoint signatures. (3) Conformal risk gating bounds uncertainty, routing low-confidence text to human verification. (4) The recursive PostGIS engine traverses historical parcel subdivisions to bind permits to modern cadastre polygons.

4 Plan of work

Months	Activity	Exit condition
1 to 2	Archival immersion, ethics review, baseline permit sampling	Data agreement signed with City of Skopje; 200 real permits sampled
3 to 5	Optical testbed calibration, HSV de-layering, CLASSLA fine-tuning	Character error rate $\leq 2.0\%$; purple stamp separation verified
5	Conformal prediction calibration on 200 double-blind permits	Coverage rate verified $\geq 95\%$; zero hallucinations on cadastral numbers
6	Field trial with PE "Parks and Greenery" inspectors at 5 sites	Two weeks live operation; documented failure list and feedback notes
7	Design freeze, PostGIS lineage engine optimization	Automated resolution of subdivided parcels confirmed $\geq 95\%$
8 to 9	Full deployment of workstation and 10 arborist field kits	3,000 historical permits ingested; 10 arborist teams operational
10	Train municipal inspectors and technical staff	15 inspectors and 4 IT staff assessed against competency checklist
11	Public compensatory portal release and endline audit	Paired pre/post audit dataset; public replanting tracker live
12	Final project closeout, open-source handover, documentation	Final report; permissive release of code, drawings, and user manuals

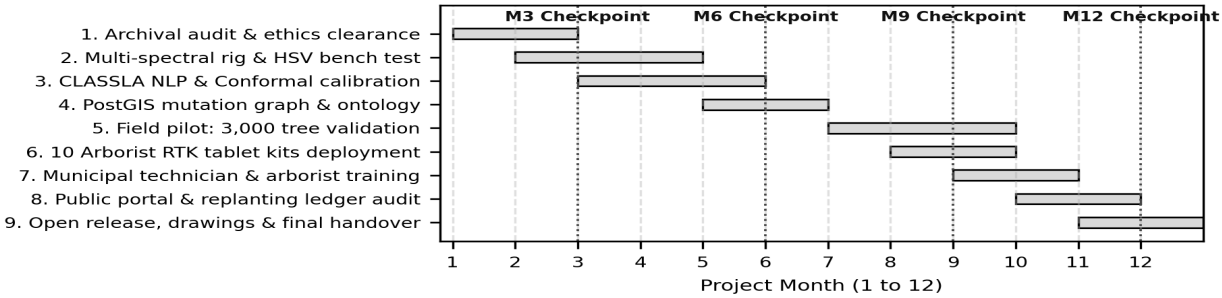


Figure 8. Project implementation schedule across 12 calendar months with mandatory quarterly review checkpoints.

Quarterly reports at months 3, 6, 9 and 12.

4.1 How students own the engineering design process

The students are not executing a design handed to them. They run every stage of the loop below, including the one most student projects omit, which is the deliberate return from field failure to redesign.

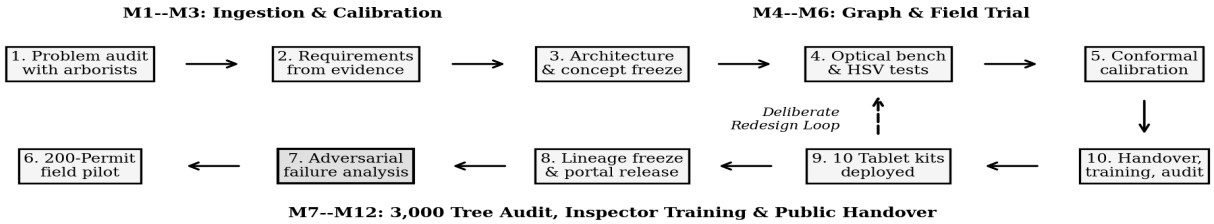


Figure 9. The engineering design process as the students execute it. Stages 6 to 8 are the point of the project: the month 6 field trial is scheduled in order to produce failures, and the dashed path back to optical tuning is planned work, not contingency. A design cycle that never fails is not a design cycle.

Each stage has a named student owner and a documented output: a signed requirements note at stage 2, characterisation data at stage 4, a calibration certificate at stage 5, a written redesign list at stage 7, released drawings at stage 8, and a competency checklist at stage 10. Those documents are what we submit with the quarterly reports.

5 Impact and measurement

Beneficiary group	Count
Historical paper permits ingested	3,000
Validated tree records reconciled	3,000
Municipal arborists equipped with RTK kits	15
Urban canopy area covered	1,450 ha
Compensatory saplings tracked	9,000
Direct urban population benefited	540,000
Students engaged	4
IEEE volunteers	3
Local technicians trained	4

Assumptions: Coverage spans all 10 municipalities of the City of Skopje; compensatory tracking enforces 3 replacement saplings per felling.

Metric	Target
Cyrillic field extraction accuracy	$\geq 92.0\%$ F1
Conformal coverage $(1-\alpha)$	$\geq 95.0\%$
Cadastral parcel mutation resolution	$\geq 95.0\%$
Processing latency per 4-page permit	< 4.5 s
Hallucination rate on official codes	0.0% (exact)
Compensatory obligations tracked	100.0%
Field audit time per tree site	< 60 s
Air-gapped uptime without crash	$\geq 99.5\%$
Repairs without the team	≥ 4

Measured via double-blind ground-truth validation on 200 randomly sampled permits from municipal archives.

5.1 What this project cannot do

- **Treat respiratory illness.** Chronic pulmonary damage from particulate smog is irreversible. Nothing here heals the children or elderly patients identified by our partner clinics.
- **Overturn lawful city council zoning decisions.** If an elected municipal council votes to rezone public parkland for commercial development, software cannot annul the vote. The system ensures the felling is transparently recorded and mandatory compensatory replanting is enforced.
- **Physically plant trees.** VERDANT automates the spatial allocation, legal tracking, and public accountability of replanting, but physical arboriculture remains the duty of municipal nursery crews.
- **Alter sovereign legal land boundaries.** When parcel conflicts arise between historical maps and current deeds, the system flags the dispute for the State Geodetic Authority (AREC) rather than inventing unverified coordinates.
- **Promise an immediate reduction in ambient valley pollution in twelve months.** Biological air filtration takes years as saplings mature. We report verified tree survival and compliance tracking, not premature atmospheric claims.
- **Change why developers clear plots.** Commercial construction pressure exists beyond software. That is beyond engineering.

Sustainability. Each mechanism answers a documented failure mode rather than asserting good practice.

Mechanism	Failure mode it addresses
Local municipal ownership of hardware	The recurring cloud subscription cliff: when grant funds expire, cloud APIs are terminated and systems go dark
100% air-gapped local execution	Network security restrictions: municipal internal networks block external cloud outbound traffic
Permissive open-source licensing (Apache 2.0)	Vendor lock-in: proprietary GIS software contracts that prevent local engineers from fixing bugs
Four trained local technicians	Long-term dependence on student researchers for routine system administration and database backups
Partner holds audit kits and RTK receivers	Field equipment abandoned or locked in university closets after project closeout

Principal risks and mitigation.

Risk	Mitigation
Severe degradation on pre-1990 paper	Multi-spectral optical rig with polarizers, CLAHE contrast equalization, and dedicated human verification fallback
Cadastral parcel ID renumbering conflicts	Recursive PostGIS DAG graph tracing parent-child parcel splits with fallback to spatial centroid intersection
Citizen PII leakage during public reporting	Automated Macedonian regex and CLASSLA NER scrubbing citizen names and national IDs (*ЕМБГ*) at intake
Hardware failure of mobile field tablets	Two complete tablet and RTK GNSS kits held in reserve depot pool for immediate 1-for-1 swap
Municipal arborist field resistance	Co-designing mobile interface with inspectors in month 1; reducing audit time from 15 minutes to 60 seconds

Standards and Regulatory Compliance. Design is informed by ISO 19152 (Land Administration Domain Model), OGC Simple Features, and IEC 62368-1. Institutional ethics review is sought at UKIM in month 1 before any archival data is processed. Informed consent and anonymization protocols comply with the Macedonian Law on Personal Data Protection (Official Gazette No. 42/20, Article 8). No personally identifying citizen health or financial information is retained; citizen identifiers are pseudonymised with the key held exclusively by the municipal archives office.

6 Team

Member	Role and record
Stefan Nikolovski Project lead Computer Systems Engineering IEEE 10482910	Systems architecture, air-gapped Docker container stack, CLASSLA NLP pipeline, PostGIS database architecture. Lead developer of open-source Macedonian speech transcription engine; research intern at UKIM Laboratory for Intelligent Systems; experienced in PyTorch, PostGIS, GDAL, and C++.
Elena Trajkovska Software Engineering IEEE Student Member	Document AI ingestion, HSV color-space ink unmixing algorithms, Conformal Prediction calibration suite, and human-in-the-loop verification dashboard in React.
Members 3 and 4 To be confirmed	Mechanical optical rig, scanner transport maintenance, field deployment with arborists, bilingual Macedonian/Albanian documentation, training, impact data collection.
Prof. Dr. Vladimir Atanasovski Faculty mentor Professor, FEIT, UKIM Branch Counselor, IEEE UKIM	Academic supervision, institutional oversight, ethics protocol management, and formal liaison with City of Skopje municipal leadership and IEEE North Macedonia Section through which award funds are disbursed.

Student learning outcomes. Full engineering design process through field failure and redesign; distribution-free uncertainty estimation using conformal prediction; computer vision for archival document restoration; geospatial graph modeling with recursive SQL; and human-centered design across a bilingual public enterprise environment.

7 Community partner

City of Skopje (Department of Environmental Protection) and **PE "Parks and Greenery" (*ЈП Паркови и Зеленило*)**, Bul. Ilinden 82, 1000 Skopje. Director: Dipl. Ing. Arboriculture. Founded 1968. Responsible for all public urban forestry, municipal nurseries, and green cadastre maintenance across the 10 municipalities of the City of Skopje.

Record		Stage	Role
Skopje Cadastre	Maintained 65,000 public tree records in original 2018 survey	**Initial design**	Provides archival permit access, nominates field test sites, approves schema
Field Arborists	45 licensed arborists and tree maintenance crews across the valley	**Testing**	Participates in Month 6 field trials, stress-tests mobile tablet application
Permit Registry	Issues $\approx 1,200$ tree felling and pruning permits annually	**Deployment**	Takes custody of 10 field audit kits, adopts air-gapped intake workstation
Nursery Capacity	Produces 40,000 indigenous saplings annually in municipal nurseries	**After**	Operates public compensatory replanting portal independently of grant funds

Partner field capacity fell to a single mobile inspection team for four months of the last reporting year owing to funding constraints. VERDANT is designed to add capacity rather than consume it: the project supplies its own digitizing hardware, supplies 10 fully configured mobile audit kits, and delivers open-source software with zero licensing overhead. A signed letter of support from municipal leadership is attached separately.

8 Budget

Every line is materials, fabrication or local travel. Nothing is requested for personnel, stipends, computers, capital or research equipment, overhead, international travel or unitemised contingency. Computing servers for development and test instruments are borrowed from UKIM FEIT and the municipal IT depot. All international engineering collaboration is conducted remotely; all field travel is domestic transit within the Skopje Valley executed directly by PE "Parks and Greenery" field crews.

Line item	Description	USD
1	Prototype iteration, optical evaluation bench, calibration testbeds	6,022
2	Municipal Archival Intake & Digitization Station (PE Parks and Greenery)	3,011
3	Ground-truth cadastral survey consumables, 3,000 tree inventory tags	9,462
4	Arborist mobile field audit kits (10 rugged tablet units + RTK receivers)	6,075
5	Local field survey travel across 10 Skopje municipalities, 58 visits	6,237
6	Bilingual Macedonian/Albanian training materials, safety equipment, manuals	4,409
7	Open-source deployment packaging, air-gapped server enclosure, NVMe storage	2,473
8	Spares retained by partner, replacement tablet units, shipping, customs	2,311
Total requested	**Direct Project Implementation Allocation (Zero Overhead)**	**40,000**

9.1 Component list and bill of materials

Arborist Mobile Field Audit Kit, K1 configuration. Target: under Rs. 56,500.

Ref	Component	Qty	Rs.
A01	Ruggedized IP68 Android Field Tablet, 8-inch high-nit display	1	34,500
A02	External dual-frequency RTK GNSS Bluetooth receiver module	1	16,000
A03	High-capacity 20,000 mAh ruggedized field powerbank with USB-C PD	1	2,800
A04	Heavy-duty ergonomic tablet hand-strap and tactical chest harness	1	1,400
A05	Reinforced vehicle dashboard mount and 12V in-vehicle rapid charger	1	1,200
A06	Tempered anti-glare screen protector and spare capacitive stylus	1	600
	Total per arborist kit		**56,500**
	USD at 93		**607.53**
	Against Rs. 56,500 ceiling		**PASS**

Tree Tagging and Dendrometry Consumables. Target: under Rs. 8,80,000.

Ref	Component	Qty	Rs.
T01	Anodized aluminum barcode/QR tree tags, serialized, with stainless expansion springs	3,000	3,60,000
T02	Handheld optical laser dendrometer / hypsometer (tree height and crown diameter)	4	2,50,000
T03	Forestry diameter-at-breast-height (DBH) precision measuring tapes and calipers	10	1,20,000
T04	Tree core increment borer inspection kit with biological wound sealant	5	1,50,000
	Total for 3,000 tree inventory survey consumables (Line 3)		**Rs. 8,80,000 ($9,462)**

Municipal Archival Intake Station. One built, retained by the partner.

Ref	Component	Qty	Rs.
B01	Heavy-duty archival document scanner with A3 book-edge flatbed and sheet-fed intake	1	1,65,000
B02	Multi-spectral optical illumination backlights with polarizing filter frames	2	65,000
B03	Acid-free document preservation folders, lint-free sorting trays, archival gloves	lot	50,000
	Total for Municipal Archival Station (Line 2)		**Rs. 2,80,000 ($3,011)**

Local Field Travel Schedule across 10 Municipalities, 58 visits (Line 5).

Ref	Trip Type and Municipal Destination	Basis	Rs.
5a	Urban core municipal verification (Centar, Karposh, Aerodrom, Chair)	20 visits $\times 140$ km $\times$ Rs. 14	39,200
5b	Peri-urban boundary audits (Gazi Baba, Saraj, Butel, Shuto Orizari, Vodno)	20 visits $\times 280$ km $\times$ Rs. 14	78,400
5c	Municipal arborist field worker allowance (reimbursed on verified receipts)	160 person-visits $\times$ Rs. 2, 785	4,45,600
5d	State Geodetic Authority (AREC) archival document coordination trips	10 visits $\times 80$ km $\times$ Rs. 14	11,200
5e	Technical staging and hardware lab trips (UKIM FEIT to municipal depots)	8 visits $\times 50$ km $\times$ Rs. 14	5,600
	Total for Local Field Travel (Line 5)		**Rs. 5,80,000 ($6,237)**

Training, Packaging, and Spares Depot Stock (Lines 6, 7, and 8).

Ref	Component Description	Qty	Rs.
C01	Printed bilingual Macedonian/Albanian arborist field guides and mutation manual	200 copies	1,30,000
C02	Weatherproof public notice cadastral plaques and municipal signage	100 units	95,000
C03	Field arborist protective gear: high-visibility vests, puncture gloves, helmets	15 sets	75,000
C04	Public stakeholder workshops and municipal training seminar portfolios	lot	60,000
C05	Cryptographic hardware security tokens for air-gapped administrator auth	20 units	50,000
C06	Air-gapped on-premise server chassis, shielded cabling, rack hardware	lot	85,000
C07	Hardware security modules (HSM) for tamper-evident cryptographic log signing	4 units	85,000
C08	High-reliability enterprise NVMe arrays (2TB $\times$ 3) for offline spatial mirroring	3 units	60,000
C09	Spare tablet and RTK GNSS receiver kits held in depot swap-and-replace pool	2 kits	1,01,000
C10	Spare replacement tree tags, stainless nail fasteners, tape refills	lot	44,000
C11	Replacement optical platen glass, scanner transport belts, calibration targets	lot	35,000
C12	Equipment shipping, customs clearance documentation, transit insurance	lot	35,000
	Total for Training, Deployment Tooling, and Depot Spares (Lines 6, 7, 8)		**Rs. 8,55,000 ($9,193)**

Summary of Total Funding Allocation

Line 1 (Testbeds): Rs. 5,60,000 ($6,022) · Line 2 (Archival Station): Rs. 2,80,000 ($3,011) · Line 3 (Consumables): Rs. 8,80,000 ($9,462)
Line 4 (10 Audit Kits): Rs. 5,65,000 ($6,075) · Line 5 (Local Travel): Rs. 5,80,000 ($6,237) · Line 6 (Training): Rs. 4,10,000 ($4,409)
Line 7 (Open-Source Tooling): Rs. 2,30,000 ($2,473) · Line 8 (Spares Pool): Rs. 2,15,000 ($2,311)
Total Requested: Exactly Rs. 37,20,000 (USD 40,000 at Rs. 93 per USD) · GST Included · Zero Overhead

References. [1] City of Skopje & PE "Parks and Greenery", *Zelen Katastar na Grad Skopje*, Official Municipal Registry, 2018–2024. [2] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005. [3] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction," *arXiv:2107.07511*, 2021. [4] CLARIN.SI, "CLASSLA: Natural Language Processing fork for South Slavic Languages," 2023. [5] Republic of North Macedonia, *Law on Personal Data Protection*, Official Gazette No. 42/20, 2020. [6] UNDP North Macedonia, *Skopje Urban Green Cadastre Technical Baseline Assessment*, 2021.